APPENDIX A
ENGINEERING DATA MANAGEMENT EXTENSION (PLM-INSPIRED)

A.1 Purpose and scope

After the static analysis of Chapters 5 and 6 was completed, the project data was organised in a small engineering data management system inspired by Product Lifecycle Management (PLM) practice. Its purpose is to keep the simulation data of the three face-sheet configurations (0.85 mm, 1.2 mm and 1.6 mm) traceable to the original ANSYS files, to check the data quality automatically, and to answer comparison questions with SQL. The system is written in Python (standard library only) around a SQLite database and is stored in the project repository next to the CAD, ANSYS and report files, none of which were modified.

> **Three kinds of data are kept apart. Source** data is read unchanged from the project files, with file and line. **Derived** data is calculated from source values (percentage changes, a stiffness proxy). **Demonstration** data (part numbers, bill of materials, revisions and a change request, Section A.8) is illustrative. This extension is not an enterprise PLM deployment and nothing in it was approved by any organisation.

A.2 Data sources and import

Table A1. Data sources used by the import

Source	Data extracted
ANSYS Workbench archives (.wbpz), one per configuration	solve.out: solver release, run date, mesh counts, mass per body, warnings. ds.dat (MAPDL input): materials, load, fixed support. file.mntr: largest displacement component. MatML.xml: material names. .wbpj: project save time.
README results table	Deformation, stress and strain per thickness (rounded values).
Report Table 6 (page 61)	Deformation, stress (Pa) and strain at full precision.
Report Tables 4 and 5, Figure 11	Material data, table labels and the 1000 N load, transcribed with page numbers and checked against the PDF text.

The import produced 3 configurations, 3 simulation cases, 98 registered files (each with its SHA-256, including every member of the archives), 61 solver inputs and boundary conditions, 33 result values and 9 results recorded as *missing*. Source files are opened read-only. The binary result file (file.rst) and the Mechanical database are registered but not parsed, so directional deformation, strain energy and volume, which exist only as contour images, are recorded as missing and are not estimated.

A.3 Database design

The database has 21 tables and one view with enforced foreign keys, uniqueness constraints and indexes on the main join columns. Every stored quantity has a canonical unit (mm, MPa, g, ...); the value and unit as printed in the source are kept next to the converted value, for example 2 989 600 Pa in Table 6 is stored as 2.9896 MPa.

Table A2. Main tables

Table(s)	Content
configuration, simulation_case, solver	Design configuration (face-sheet thickness, core and face-sheet material), its analysis, and the solver version.
simulation_input	Analysis settings, mesh, load, supports and material assignment; derived rows carry their formula.
result_metric, material_property	Results and material data with value, unit, printed value, status (ok / missing) and origin (solver file, documented, derived). Solver and report material data are stored side by side.
source_file, artifact_link	Every file with SHA-256 and the role it plays for each configuration (input deck, solver log, contour image, ...).
part, bom_line	Bill of materials per configuration (demonstration).
design_revision, change_request, ecr_item, ecr_status_history	Design revisions, change request, affected items, verification cases and status history (demonstration).
validation_run, validation_issue	Data-quality findings with severity, affected record and suggested correction.

A.4 Traceability

Each imported value can be followed to the file, archive member and line it came from. Table A3 shows this for configuration T085 (0.85 mm).

Table A3. Traceability of selected values, configuration T085 (0.85 mm)

Item	Value	Source file	Locator	SHA-256 (first 12)
total_deformation	0.0070107 mm	MAJOR PROJECT REPORT (1).pdf	report p.61 Table 6 row 0.85	3a80aae3a369
equivalent_stress	2.9896 MPa	MAJOR PROJECT REPORT (1).pdf	report p.61 Table 6 row 0.85	3a80aae3a369
total_mass	41.041 g	0.85mm archive: solve.out	line 493	ed9a991bc759
max_displacement_component	-0.0068005 mm	0.85mm archive: file.mntr	line 11	a246ea634d37
applied pressure	-0.144251935 MPa	0.85mm archive: ds.dat	line 20372	b51b3dc9d758
applied force (derived)	1000.000 N	0.85mm archive: ds.dat	|pressure| x loaded area (derived)	b51b3dc9d758

A.5 Solver inputs recovered from the ANSYS files

Table A4. Inputs read from the solver files (identical for the three cases)

Parameter	Value	Origin
Solver / analysis	ANSYS MAPDL 2025 R1 (build 25.1), static structural	solve.out
Mesh	14,009 nodes, 2,093 solid elements (same for all cases)	solve.out
PLA core	E = 3500 MPa, ν = 0.36, ρ = 1.25 g/cm³	ds.dat
ePA-CF face sheets	E = 9000 MPa, ν = 0.35, ρ = 1.15 g/cm³	ds.dat
Support	all translational degrees of freedom fixed on 642 nodes	ds.dat
Load	pressure 0.144252 MPa on the loaded face (surface-effect elements)	ds.dat
Loaded area / resultant force	6932.316 mm² / 1000.000 N (derived: sum of face areas; |pressure| × area)	derived

The resultant force derived from the deck, 1000.000 N, equals the 1000 N load shown in Figure 11 of the report. Total mass is 41.041 g, 46.621 g and 52.999 g for 0.85, 1.2 and 1.6 mm. The core mass is 27.488 g in every case, and the face-sheet thickness implied by the solver mass, m/(ρA), is 0.850, 1.200 and 1.600 mm, which agrees with the thickness in each folder name. Solver, units, load, support, mesh counts and material data are identical in the three cases, so only the face-sheet thickness varies.

A.6 Results

Table A5. Results by face-sheet thickness (source: report Table 6; percentage changes derived)

Thickness (mm)	Total deformation (mm)	Equivalent stress (MPa)	Equivalent elastic strain	Deformation vs 0.85 mm	Stress vs 0.85 mm
0.85	0.0070107	2.9896	0.00089581	-	-
1.2	0.006009	2.9189	0.00087613	-14.29 %	-2.36 %
1.6	0.0055584	2.8536	0.00085723	-20.72 %	-4.55 %

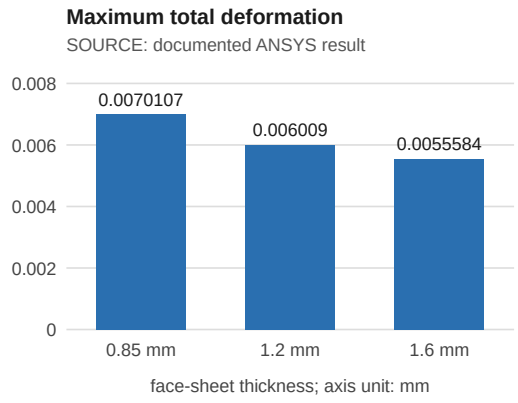


Figure A1. Maximum total deformation (source)

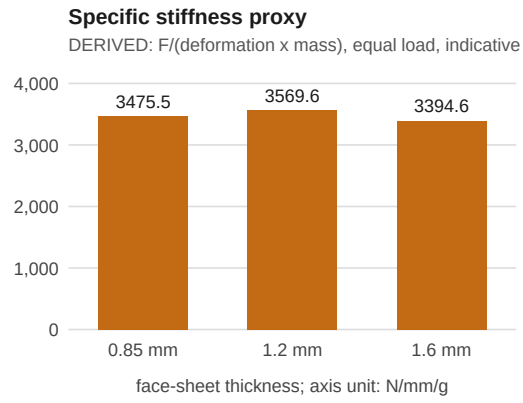


Figure A2. Specific stiffness proxy (derived)

Table A6. Ranking and derived stiffness per mass (equal load; F/δ with δ = maximum total deformation)

Thickness (mm)	Mass (g)	Stiffness F/δ (N/mm)	Specific stiffness (N/mm/g)	Change vs 0.85 mm	Rank: deformation / specific stiffness
0.85	41.041	142,639	3475.5	-	3 / 2
1.2	46.621	166,417	3569.6	+2.7 %	2 / 1
1.6	52.999	179,908	3394.6	-2.3 %	1 / 3

Deformation, stress and strain are lowest for the 1.6 mm face sheet, and the 0.85 mm panel is the lightest. The derived stiffness per unit mass is highest at 1.2 mm and is 2.3 % lower than the 0.85 mm value at 1.6 mm, so the statement that thicker face sheets are more efficient depends on the metric: they are stiffer, but the added mass grows almost as fast as the stiffness. The proxy assumes equal load and uses the maximum deformation, not the deflection at the load point, so it is indicative only.

A.7 Data-quality checks

42 check codes test for missing values, duplicate identifiers, wrong data types, inconsistent units, invalid geometry, material and load parameters, broken links between configurations, cases and results, comparability of the cases, and consistency between the report, the README and the solver files. Each finding is classified as error, warning or information and lists the affected record and a suggested correction. Source data is never changed. On this data the checks report **0 errors, 11 warnings and 14 information items**. The warnings are listed in Table A7.

Table A7. Findings reported by the data-quality checks (none were corrected in the source files)

Finding	Detail	Suggested correction
Material data, report vs solver	Table 4 gives flexural modulus 1973 / 5160 MPa and density 1.23 / 1.24 g/cm³ (PLA / ePA-CF). The solver used E = 3500 / 9000 MPa and density 1.25 / 1.15 g/cm³.	State the source of the solver material data and align Table 4 with it.
Table 5 labels	Core is labelled “PLA -CF” and face sheet “PLA”; the solver has a PLA core and ePA-CF face sheets.	Correct the labels.
Table 6 unit	The equivalent elastic strain column is labelled (mm); strain is dimensionless.	Remove the unit.
Duplicate project file	The loose 0.85mm Workbench project file is byte-identical to the one inside the 1.6 mm archive.	Replace it with the project file of the 0.85 mm case.
README claim	“Improved stiffness-to-weight ratio significantly” is not supported by the derived metric of Table A6.	Define the metric or reword the claim.

Checks that confirm consistency passed: the README values equal the report values rounded to the README precision; the displacement component from the solver monitor lies in the mathematically possible range for the reported total deformation; the thickness implied by solver mass matches each configuration; and the load derived from the solver deck matches the report. These checks show internal consistency only. The repository contains no

experimental data and no mesh-convergence study, so the physical validity of the results is not established by them.

A.8 Revision and change-management demonstration

DEMONSTRATION DATA. Part numbers carry the prefix DEMO-. The bill of materials mirrors the three bodies of the ANSYS model. The revisions and the change request are a simulated workflow linked to the real simulation results; they are not records of a real PLM system and no approval by any organisation is implied.

Table A8. Demonstration bill of materials (per assembly)

Configuration	Part number	Description	Quantity	Material
T085	DEMO-HSP-2000	Hexagonal honeycomb core	1	PLA
T085	DEMO-HSP-3085	Face sheet 0.85 mm	2	ePA-CF
T120	DEMO-HSP-3120	Face sheet 1.2 mm	2	ePA-CF
T160	DEMO-HSP-3160	Face sheet 1.6 mm	2	ePA-CF

The core part DEMO-HSP-2000 is common to all configurations. Revision A is the 0.85 mm baseline (T085). Revision B (T160, 1.6 mm) was created by the demonstration change request **DEMO-ECR-0001**, “Increase face-sheet thickness from 0.85 mm to 1.6 mm”, whose status history is DRAFT → SUBMITTED → UNDER_REVIEW → IMPLEMENTED → VERIFIED. The change request lists the affected and proposed parts and configurations and links the verification simulation cases SIM-T085 and SIM-T160, whose documented maximum total deformation is 0.0070107 mm and 0.0055584 mm (-20.7 %, derived). The workflow enforces rules such as: an illegal status jump is rejected, a change request cannot be marked implemented without a revision, and it cannot be marked verified without linked simulation results.

A.9 Limitations

- One solver (ANSYS MAPDL 2025 R1) and one run per configuration exist, so no cross-solver comparison is possible; the corresponding query correctly returns no rows.
- Raw nodal results and the SpaceClaim geometry are not parsed; directional deformation, strain energy and volume are missing.
- Report values were transcribed once from the PDF and verified by text search on the cited page.
- All cases use one mesh and there is no mesh-convergence study, so mesh sensitivity cannot be assessed.
- Stiffness-based metrics are proxies (equal load, maximum deformation).

A.10 Reproducibility

The whole analysis is reproduced from the repository with *python -m plm all* (initialise, import, load demonstration data, validate, report). The command writes the engineering report, the data-quality report, the SQL query outputs and the charts to the folder *generated_reports*. Documented SQL queries are in the folder *sql*, and 78 automated tests, including deliberately injected data errors, are run with *python -m unittest discover*.